Gaussian Pulse Propagation in Optical Fibers: A Complete, Intuitive Explanation

Step-by-Step Derivation with Physical Insight

February 3, 2026

Contents

1 The Big Picture: What Happens to a Pulse? 2
- 1.1 The Question . . . 2
- 1.2 Why Gaussian Pulses? . . . 3

2 Starting Point: The Input Pulse 3
- 2.1 Mathematical Form . . . 3
- 2.2 Physical Picture . . . 3
- 2.3 Pulse Width Definition . . . 4

3 The Key Tool: Fourier Transform 4
- 3.1 Why Use Fourier Transform? . . . 4
- 3.2 Fourier Transform of Gaussian . . . 4
- 3.3 Physical Interpretation . . . 4

4 Dispersion: The Taylor Series 5
- 4.1 What is Dispersion? . . . 5
- 4.2 Taylor Expansion . . . 5
- 4.3 Physical Meaning of Each Term . . . 5
- 4.4 Why β_2 Causes Broadening . . . 5

5 Propagation in Frequency Domain 6
- 5.1 Propagation Formula . . . 6
- 5.2 Substituting the Gaussian . . . 6

6 The Magic: Completing the Square 6
- 6.1 Combining Terms . . . 6
- 6.2 Completing the Square . . . 7

7 Inverse Fourier Transform: The Gaussian Integral **7**
7.1 The Integral . . . 7
7.2 Solving the Integral . . . 7
7.3 The Final Amplitude Expression . . . 8

8 Pulse Width Evolution: The Key Result **8**
8.1 Separating Real and Imaginary Parts . . . 8
8.2 Dispersion Length and Final Width . . . 9
8.3 Physical Interpretation . . . 9

9 Complete Picture: Delay + Broadening + Loss **9**
9.1 Three Effects Combined . . . 9
9.2 Summary of Effects . . . 10

10 Worked Example: Real Numbers **10**
10.1 Problem Setup . . . 10
10.2 Solution Step-by-Step . . . 10
10.3 Visualization . . . 11

11 Physical Insights **12**
11.1 Why Does the Pulse Broaden? . . . 12
11.2 Sign of β_2 . . . 12
11.3 Chirp: Frequency Sweep . . . 12

12 Engineering Applications **12**
12.1 Dispersion Compensation . . . 12
12.2 Bit Rate Limits . . . 13
12.3 Dispersion-Bit Rate Product . . . 13

13 Connection to Quantum Communication **13**

14 Summary: Key Equations **14**

1 The Big Picture: What Happens to a Pulse?

1.1 The Question

When you send a short laser pulse (like 10 picoseconds) through an optical fiber, what happens to it? **Answer:** Three main effects:

1. **Delay**: The pulse takes time to propagate (travels at group velocity)
2. **Broadening**: The pulse spreads out in time (due to dispersion)
3. **Attenuation**: The pulse weakens (due to loss)

1.2 Why Gaussian Pulses?

Gaussian pulses are important because:

- Many lasers naturally produce Gaussian-shaped pulses
- Mathematically tractable (Gaussian $\leftrightarrow$ Gaussian under Fourier transform)
- Good model for real pulse shapes
- Preserves shape during linear propagation (only width changes)

2 Starting Point: The Input Pulse

2.1 Mathematical Form

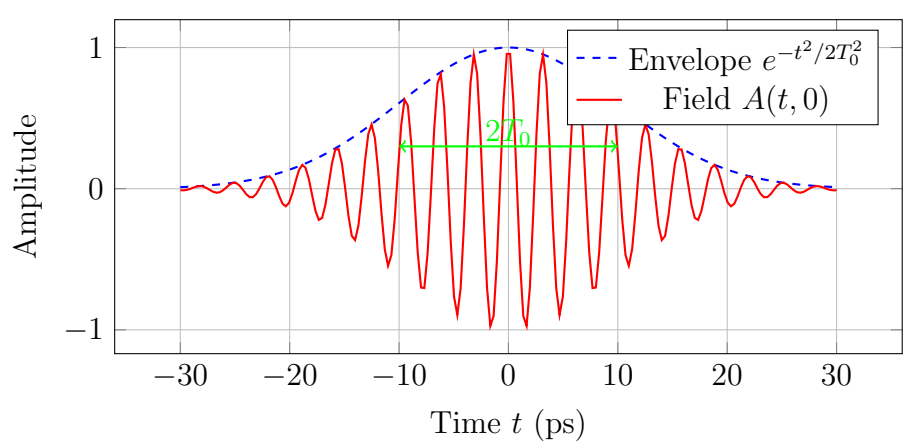
At the fiber input ($z = 0$), we have a Gaussian pulse:

$$\boxed{A(t,0) = A_0 \exp\left(-\frac{t^2}{2T_0^2}\right) e^{i\omega_0 t}} \tag{1}$$

Let's understand each part:

- A_0 = peak amplitude (V/m or similar)
- $\exp(-t^2/2T_0^2)$ = Gaussian envelope (bell curve)
- T_0 = **initial pulse width** (1/e half-width in amplitude)
- $e^{i\omega_0 t}$ = fast oscillation at carrier frequency ω_0

2.2 Physical Picture



2.3 Pulse Width Definition

The **full width at half maximum (FWHM)** is:

$$\text{FWHM} = 2\sqrt{2\ln 2}\,T_0 \approx 1.665\,T_0 \tag{2}$$

So if $T_0 = 10$ ps, FWHM ≈ 16.65 ps.

3 The Key Tool: Fourier Transform

3.1 Why Use Fourier Transform?

The propagation equation is simple in the **frequency domain**:

$$\tilde{A}(\omega, z) = \tilde{A}(\omega, 0) \cdot e^{i\beta(\omega)z}$$

Each frequency component just picks up a phase $e^{i\beta(\omega)z}$! In the time domain, this would be a complicated convolution. Fourier transforms make it simple.

3.2 Fourier Transform of Gaussian

Starting with:

$$A(t, 0) = A_0 e^{-t^2/2T_0^2} e^{i\omega_0 t}$$

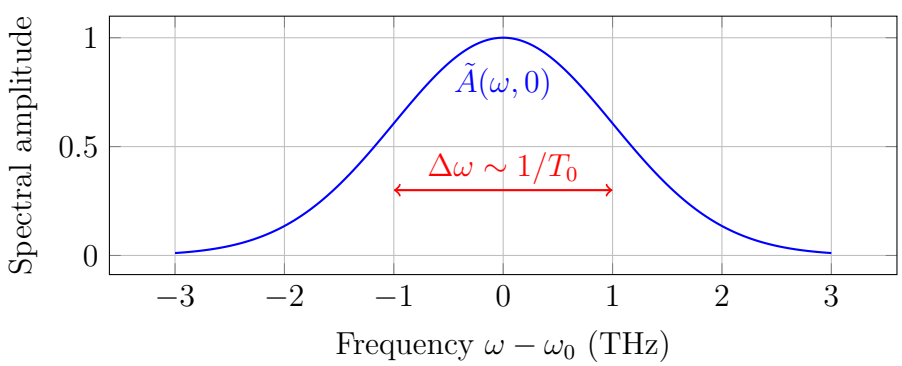
The Fourier transform is:

$$\tilde{A}(\omega, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(t, 0) e^{i\omega t} dt \tag{3}$$

Key fact: Fourier transform of Gaussian is Gaussian!

$$\boxed{\tilde{A}(\omega, 0) = A_0 T_0 \sqrt{2\pi} \exp\left(-\frac{(\omega - \omega_0)^2 T_0^2}{2}\right)} \tag{4}$$

3.3 Physical Interpretation



Time-bandwidth product:

$$\Delta t \cdot \Delta\omega \sim 1$$

Shorter pulse (small T_0) $\Rightarrow$ broader spectrum (large $\Delta\omega$)!

4 Dispersion: The Taylor Series

4.1 What is Dispersion?

Different frequencies travel at different speeds! The propagation constant $\beta(\omega)$ tells us how fast each frequency propagates:

$$\beta(\omega) = \text{phase accumulated per unit length at frequency } \omega \tag{5}$$

4.2 Taylor Expansion

Expand $\beta(\omega)$ around the carrier frequency ω_0:

$$\boxed{\beta(\omega) = \beta_0 + \beta_1(\omega - \omega_0) + \frac{\beta_2}{2}(\omega - \omega_0)^2 + \frac{\beta_3}{6}(\omega - \omega_0)^3 + \cdots} \tag{6}$$

4.3 Physical Meaning of Each Term

$$\beta_0 = \beta(\omega_0) \quad \text{Overall phase (unimportant for pulse shape)} \tag{7}$$

$$\beta_1 = \left.\frac{d\beta}{d\omega}\right|_{\omega_0} = \frac{1}{v_g} \quad \text{Inverse group velocity (delay)} \tag{8}$$

$$\beta_2 = \left.\frac{d^2\beta}{d\omega^2}\right|_{\omega_0} \quad \text{Group velocity dispersion (GVD) (broadening!)} \tag{9}$$

$$\beta_3 = \left.\frac{d^3\beta}{d\omega^3}\right|_{\omega_0} \quad \text{Third-order dispersion (pulse asymmetry)} \tag{10}$$

4.4 Why β_2 Causes Broadening

The β_2 term means:

$$\text{phase} \sim \frac{\beta_2}{2}(\omega - \omega_0)^2 z$$

Different frequencies acquire **different phases**!

- Red frequencies ($\omega < \omega_0$): one phase
- Blue frequencies ($\omega > \omega_0$): different phase

When we inverse Fourier transform, these phase differences cause the pulse to spread out.

5 Propagation in Frequency Domain

5.1 Propagation Formula

After propagating distance z:

$$\tilde{A}(\omega, z) = \tilde{A}(\omega, 0) \cdot e^{i\beta(\omega)z} \cdot e^{-\alpha z/2} \tag{11}$$

Keeping only up to β_2:

$$\tilde{A}(\omega, z) = \tilde{A}(\omega, 0) \cdot \exp\left[i\beta_0 z + i\beta_1(\omega - \omega_0)z + \frac{i\beta_2}{2}(\omega - \omega_0)^2 z\right] \tag{12}$$

5.2 Substituting the Gaussian

$$\begin{aligned}\tilde{A}(\omega, z) = A_0 T_0 \sqrt{2\pi} \exp\left[-\frac{(\omega - \omega_0)^2 T_0^2}{2}\right] \\ \times \exp\left[i\beta_0 z + i\beta_1(\omega - \omega_0)z + \frac{i\beta_2}{2}(\omega - \omega_0)^2 z\right]\end{aligned} \tag{13}$$

Combining the exponentials:

$$\tilde{A}(\omega, z) = A_0 T_0 \sqrt{2\pi} e^{i\beta_0 z} \exp\left[-\frac{(\omega - \omega_0)^2 T_0^2}{2} + i\beta_1(\omega - \omega_0)z + \frac{i\beta_2 z}{2}(\omega - \omega_0)^2\right] \tag{14}$$

6 The Magic: Completing the Square

6.1 Combining Terms

Define $\Omega = \omega - \omega_0$ for simplicity. The exponent in the equation above is:

$$-\frac{\Omega^2 T_0^2}{2} + i\beta_1 \Omega z + \frac{i\beta_2 z}{2}\Omega^2 \tag{15}$$

Factor out the Ω^2 terms:

$$= -\frac{1}{2}\left(T_0^2 - i\beta_2 z\right)\Omega^2 + i\beta_1 z\Omega \tag{16}$$

Define the complex pulse width parameter:

$$T^2(z) = T_0^2 - i\beta_2 z \tag{17}$$

Substituting this back, we get a concise quadratic form (note that $T^2(z)$ is in the numerator):

$$\boxed{\text{Exponent} = -\frac{1}{2}T^2(z)\Omega^2 + i\beta_1 z\Omega} \tag{18}$$

6.2 Completing the Square

We want to write this quadratic in the standard form $-a(\Omega - k)^2 + C$ to prepare it for integration.

Standard form: $-a\Omega^2 + b\Omega = -a\left(\Omega - \frac{b}{2a}\right)^2 + \frac{b^2}{4a}$

Identifying our coefficients:

- $a = \frac{T^2(z)}{2}$
- $b = i\beta_1 z$

Applying the standard form:

$$\text{Square Term} = -\frac{T^2(z)}{2}\left(\Omega - \frac{i\beta_1 z}{T^2(z)}\right)^2 \tag{19}$$

And the constant term (which comes out of the square):

$$\text{Constant} = \frac{b^2}{4a} = \frac{(i\beta_1 z)^2}{4(T^2(z)/2)} = \frac{-\beta_1^2 z^2}{2T^2(z)} \tag{20}$$

Thus, the full exponent is:

$$-\frac{T^2(z)}{2}\left(\Omega - \frac{i\beta_1 z}{T^2(z)}\right)^2 - \frac{\beta_1^2 z^2}{2T^2(z)} \tag{21}$$

7 Inverse Fourier Transform: The Gaussian Integral

7.1 The Integral

We now perform the inverse Fourier transform to get back to the time domain:

$$A(t,z) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \tilde{A}(\omega,z)e^{-i\omega t}d\omega \tag{22}$$

Substituting $\omega = \Omega + \omega_0$ and our completed square form:

$$A(t,z) = \frac{A_0 T_0\sqrt{2\pi}}{\sqrt{2\pi}}e^{i(\beta_0 z - \omega_0 t)}e^{-\frac{\beta_1^2 z^2}{2T^2(z)}}\int_{-\infty}^{\infty}\exp\left[-\frac{T^2(z)}{2}\left(\Omega - \frac{i\beta_1 z}{T^2(z)}\right)^2\right]e^{-i\Omega t}d\Omega \tag{23}$$

7.2 Solving the Integral

This integral is the Fourier transform of a shifted Gaussian. We can solve it by shifting the integration variable $u = \Omega - \frac{i\beta_1 z}{T^2(z)}$.

The standard Gaussian integral result is:

$$\int_{-\infty}^{\infty} e^{-Au^2}e^{-iut}du = \sqrt{\frac{\pi}{A}}e^{-t^2/4A} \tag{24}$$

Here, $A = T^2(z)/2$. The result of the integration is:

$$\text{Integral} = \sqrt{\frac{2\pi}{T^2(z)}} \exp\left[-\frac{(t-\beta_1 z)^2}{2T^2(z)}\right] \tag{25}$$

(Note: The $(t-\beta_1 z)$ term arises from combining the external constant term and the Fourier phase shift.)

7.3 The Final Amplitude Expression

Combining all prefactors and exponents:

$$\boxed{A(t,z) = \frac{A_0 T_0}{\sqrt{T^2(z)}} \exp\left[-\frac{(t-\beta_1 z)^2}{2T^2(z)}\right] e^{i(\beta_0 z - \omega_0 t)}} \tag{26}$$

Substituting $T^2(z) = T_0^2(1 - i\beta_2 z/T_0^2)$:

$$A(t,z) = \frac{A_0}{\sqrt{1 - i\beta_2 z/T_0^2}} \exp\left[-\frac{(t-\beta_1 z)^2}{2T_0^2(1 - i\beta_2 z/T_0^2)}\right] e^{i(\beta_0 z - \omega_0 t)} \tag{27}$$

8 Pulse Width Evolution: The Key Result

8.1 Separating Real and Imaginary Parts

The exponent governing the pulse shape is:

$$-\frac{(t-\beta_1 z)^2}{2T^2(z)} \tag{28}$$

To find the physical pulse width, we must extract the real part of $1/T^2(z)$:

$$\frac{1}{T^2(z)} = \frac{1}{T_0^2 - i\beta_2 z} = \frac{T_0^2 + i\beta_2 z}{(T_0^2 - i\beta_2 z)(T_0^2 + i\beta_2 z)} = \frac{T_0^2 + i\beta_2 z}{T_0^4 + \beta_2^2 z^2} \tag{29}$$

The real part of the exponent is therefore:

$$\text{Re}\left[-\frac{(t-\beta_1 z)^2}{2T^2(z)}\right] = -\frac{(t-\beta_1 z)^2}{2} \cdot \frac{T_0^2}{T_0^4 + \beta_2^2 z^2} \tag{30}$$

We rewrite this in the standard Gaussian form $-\frac{(t-\beta_1 z)^2}{2T(z)^2}$:

$$\frac{1}{T(z)^2} = \frac{T_0^2}{T_0^4 + \beta_2^2 z^2} \implies T(z)^2 = \frac{T_0^4 + \beta_2^2 z^2}{T_0^2} \tag{31}$$

8.2 Dispersion Length and Final Width

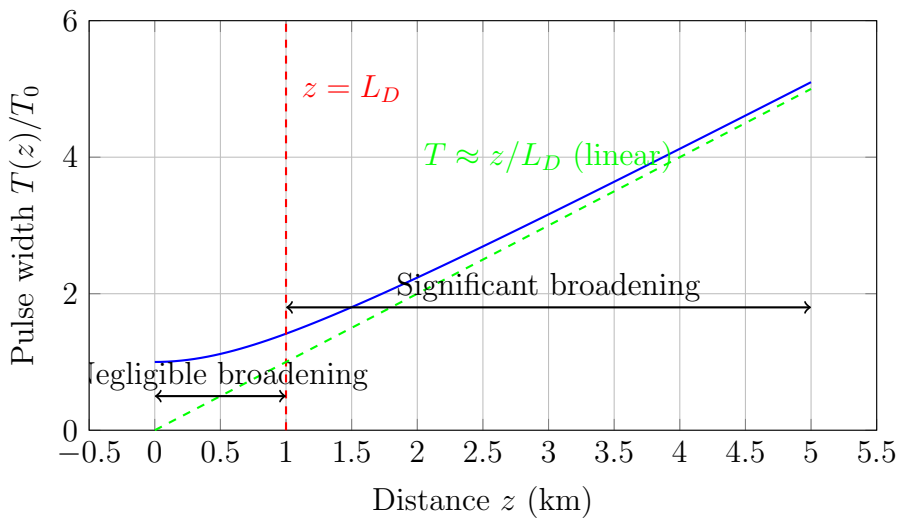
Dividing through by T_0^2:

$$T(z)^2 = T_0^2 \left(1 + \frac{\beta_2^2 z^2}{T_0^4}\right) \tag{32}$$

Defining the **Dispersion Length** $L_D = \frac{T_0^2}{|\beta_2|}$, we get the famous broadening formula:

$$\boxed{T(z) = T_0 \sqrt{1 + \left(\frac{z}{L_D}\right)^2}} \tag{33}$$

8.3 Physical Interpretation



Three regimes:

1. $z \ll L_D$**:** Negligible broadening
$$T(z) \approx T_0$$

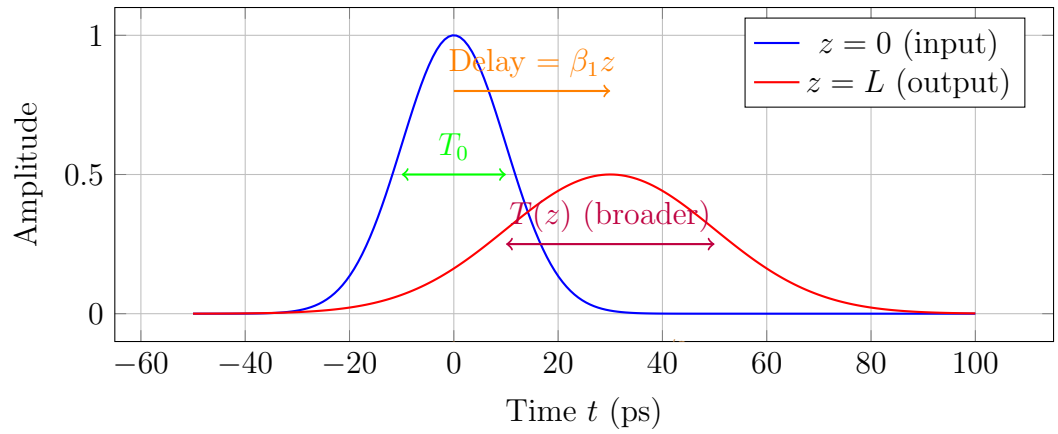
2. $z = L_D$**:** $\sqrt{2}$ broadening (41% increase)
$$T(L_D) = \sqrt{2} T_0$$

3. $z \gg L_D$**:** Linear broadening
$$T(z) \approx \frac{|\beta_2| z}{T_0}$$

9 Complete Picture: Delay + Broadening + Loss

9.1 Three Effects Combined

$$\boxed{A(t,z) = \frac{A_0 e^{-\alpha z/2}}{\sqrt{1 - i\beta_2 z/T_0^2}} \exp\left[-\frac{(t - \beta_1 z)^2}{2T_0^2(1 - i\beta_2 z/T_0^2)}\right] e^{i(\beta_0 z - \omega_0 t)}} \tag{34}$$



9.2 Summary of Effects

Effect	Parameter	Result
Delay	$\beta_1 = 1/v_g$	Peak shifts to $t = \beta_1 z$
Broadening	β_2 (GVD)	Width increases: $T(z) = T_0\sqrt{1 + (z/L_D)^2}$
Attenuation	α	Amplitude decreases: $A \to Ae^{-\alpha z/2}$

10 Worked Example: Real Numbers

10.1 Problem Setup

A fiber with:

- Length: $L = 10$ km
- Core index: $n_{\text{core}} = 1.45$
- GVD: $\beta_2 = -20$ ps^2/km
- Attenuation: $\alpha = 0.2$ dB/km
- Input pulse width: $T_0 = 10$ ps
- Wavelength: $\lambda_0 = 1550$ nm

10.2 Solution Step-by-Step

(a) Propagation delay:

$$\Delta t = \beta_1 L = \frac{n_{\text{core}} L}{c} \tag{35}$$

$$= \frac{1.45 \times 10^4 \text{ m}}{3 \times 10^8 \text{ m/s}} \tag{36}$$

$$= 4.833 \times 10^{-5} \text{ s} = \boxed{48.33\ \mu\text{s}} \tag{37}$$

(b) Dispersion length:

$$L_D = \frac{T_0^2}{|\beta_2|} = \frac{(10\ \text{ps})^2}{20\ \text{ps}^2/\text{km}} \tag{38}$$

$$= \boxed{5\ \text{km}} \tag{39}$$

(c) Output pulse width:

$$T(L) = T_0\sqrt{1 + \left(\frac{L}{L_D}\right)^2} \tag{40}$$

$$= 10\sqrt{1 + \left(\frac{10}{5}\right)^2} \tag{41}$$

$$= 10\sqrt{1+4} = 10\sqrt{5} \tag{42}$$

$$= \boxed{22.36\ \text{ps}} \tag{43}$$

Broadening factor: $T(L)/T_0 = 2.236$ (124% increase!)

(d) Amplitude attenuation:

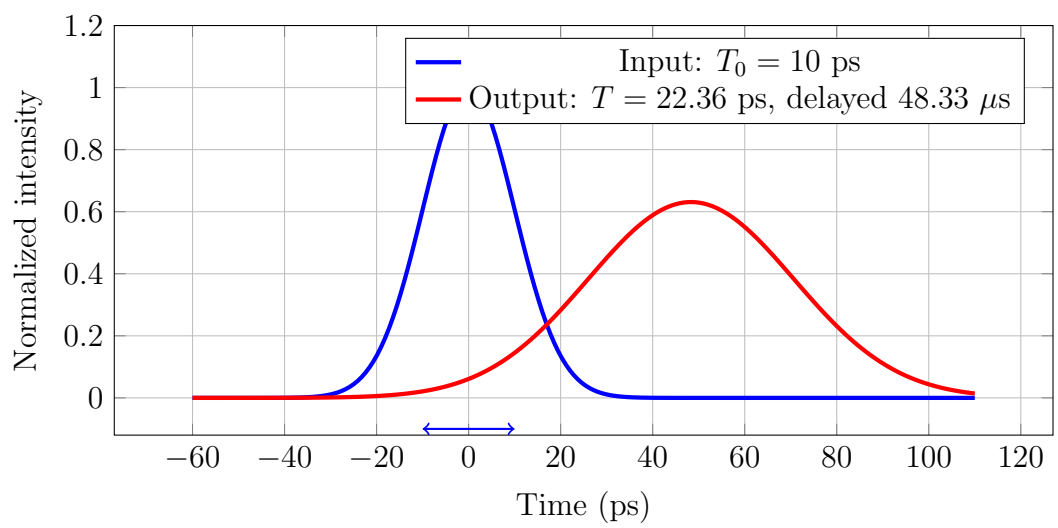
$$\alpha_{\text{Np}} = 0.1151 \times 0.2 = 0.02302\ \text{Np/km} \tag{44}$$

$$A(L) = A_0 e^{-\alpha_{\text{Np}} L} = A_0 e^{-0.2302} \tag{45}$$

$$= \boxed{0.794\, A_0} \tag{46}$$

Power: $P(L) = P_0 \times (0.794)^2 = 0.631 P_0$ (63.1% remains)

10.3 Visualization



11 Physical Insights

11.1 Why Does the Pulse Broaden?

Think of the pulse as containing many frequency components:

1. **Red components** ($\omega < \omega_0$): travel slower (if $\beta_2 < 0$)
2. **Blue components** ($\omega > \omega_0$): travel faster (if $\beta_2 < 0$)

Initially, all components are synchronized (constructive interference at $t = 0$). After propagation, they're out of sync $\rightarrow$ pulse spreads!

11.2 Sign of β_2

- $\beta_2 < 0$ **(normal dispersion):** Blue faster than red
- $\beta_2 > 0$ **(anomalous dispersion):** Red faster than blue
- $\beta_2 = 0$**:** Zero-dispersion wavelength (no broadening!)

For standard silica fiber:

- $\lambda < 1.3$ μm: $\beta_2 < 0$ (normal)
- $\lambda = 1.3$ μm: $\beta_2 = 0$ (zero-dispersion)
- $\lambda > 1.3$ μm: $\beta_2 > 0$ (anomalous)

11.3 Chirp: Frequency Sweep

The output pulse has a **chirp** - instantaneous frequency varies with time! For $\beta_2 < 0$:

- Leading edge: lower frequency (red)
- Trailing edge: higher frequency (blue)

This is why you can compress a chirped pulse with a grating!

12 Engineering Applications

12.1 Dispersion Compensation

To transmit over long distances without broadening:

1. **Dispersion-shifted fiber:** Design for $\beta_2 = 0$ at operating wavelength
2. **Dispersion compensation:** Use fiber with opposite β_2
3. **Pre-chirping:** Chirp pulse before transmission so dispersion compresses it

12.2 Bit Rate Limits

For digital communication, pulses must not overlap. If pulse width after distance L is $T(L)$, bit rate is limited by:

$$B < \frac{1}{2T(L)} = \frac{1}{2T_0\sqrt{1+(L/L_D)^2}} \tag{47}$$

For $L \gg L_D$:

$$B < \frac{T_0}{2|\beta_2|L} \tag{48}$$

12.3 Dispersion-Bit Rate Product

A key figure of merit:

$$B \cdot L < \frac{1}{2|\beta_2|B} \tag{49}$$

Example: For $\beta_2 = -20$ ps^2/km and $B = 10$ Gb/s:

$$L < \frac{1}{2 \times 20 \times 10^{-24} \times 10^{10}} = 2500 \text{ km} \tag{50}$$

13 Connection to Quantum Communication

In quantum communication, these same effects apply to quantum states:

$$|\alpha(0)\rangle \xrightarrow{\text{propagate}} |\alpha(L)\rangle \tag{51}$$

where:

$$\alpha(L) = \alpha(0)e^{i\beta(\omega)L}e^{-\alpha L/2} \tag{52}$$

The dispersion causes:

- Phase uncertainty
- Timing jitter
- Reduced distinguishability of quantum states

All affecting channel capacity!

14 Summary: Key Equations

$$\text{Input:} \quad A(t,0) = A_0 e^{-t^2/2T_0^2} e^{i\omega_0 t} \tag{53}$$

$$\text{Dispersion length:} \quad L_D = \frac{T_0^2}{|\beta_2|} \tag{54}$$

$$\text{Output width:} \quad T(z) = T_0\sqrt{1+(z/L_D)^2} \tag{55}$$

$$\text{Delay:} \quad \Delta t = \beta_1 z = \frac{n_{\text{eff}} z}{c} \tag{56}$$

$$\text{Attenuation:} \quad A \to A e^{-\alpha z/2} \tag{57}$$

$$\text{Complete output:} \quad A(t,z) = \frac{A_0 e^{-\alpha z/2}}{\sqrt{1 - i\beta_2 z/T_0^2}} \exp\left[-\frac{(t-\beta_1 z)^2}{2T_0^2(1-i\beta_2 z/T_0^2)}\right] e^{i(\beta_0 z - \omega_0 t)} \tag{58}$$